Chapter 37: Orthogonality

Because the CBS inequality holds for all inner product spaces:

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \, \|\mathbf{w}\|$$

Then if $\mathbf{u}, \mathbf{v}$ are two nonzero vectors in an inner product space, we have

$$-1 \leq \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \, \|\mathbf{v}\|} \leq 1.$$

This means that there is a unique angle $0 \leq \theta \leq \pi$, such that

$$\cos\theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

Definition: The **angle** between nonzero vectors $\mathbf{u}, \mathbf{v}$ in an inner product space is θ if and only if

$$\cos\theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

Example: For the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle = u_1v_1 + 3u_2v_2$ on $\mathbb{R}^2$, find the angle between $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$\cos\theta = \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} = \frac{u_1v_1 + 3u_2v_2}{\sqrt{u_1^2 + 3u_2^2} \cdot \sqrt{v_1^2 + 3v_2^2}} = \frac{3}{2 \cdot \sqrt{3}}$$

$$= \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}.$$

Definition: Two vectors **u** and **v** in an inner product space are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Example: P_1 is an inner product space with the following inner product:

$$\langle p, q \rangle = \int_0^1 p(x)q(x)\mathrm{d}x.$$

Are x and $x - \frac{2}{3}$ orthogonal polynomials in P_1?

$$\langle x, x - \tfrac{2}{3} \rangle = \int_0^1 x(x - \tfrac{2}{3})\,dx$$

$$= \left(\tfrac{1}{3}x^3 - \tfrac{1}{3}x^2\right)\Big|_0^1$$

$$= 0$$

$\Rightarrow$ they are orthogonal.

Example: Consider M_{22}, with the inner product $\langle A, B \rangle = \mathrm{tr}(AB^T)$. Find a value for x so that $A = \begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 2 \\ x & 0 \end{bmatrix}$ are orthogonal.

$$\langle A, B \rangle = \mathrm{tr}\left(\begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix} \cdot \begin{bmatrix} -3 & x \\ 2 & 0 \end{bmatrix} \right)$$

$$= \mathrm{tr}\left(\begin{bmatrix} -3 & x \\ 10 & -2x \end{bmatrix} \right)$$

$$= -3 - 2x = 0 \Rightarrow x = \frac{-3}{2}$$

Definition: A set S of vectors in an inner product space V is called **orthogonal** if every pair of distinct vectors in S are **orthogonal**.

Example: Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ be three vectors in $\mathbb{R}^3$. Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ an orthogonal set (with respect to the dot product)?

$v_1 \cdot v_2 = -2 + 2 = 0,$

$v_1 \cdot v_3 = 0$

$v_2 \cdot v_3 = 0$

$\Rightarrow \{v_1, v_2, v_3\}$ is an orthogonal set.

Definition: If each vector of an orthogonal set S has unit length, then S is called **orthonormal**.

Example: Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ be three vectors in $\mathbb{R}^3$. Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ an orthonormal set? If not, make an orthonormal set using this set as a starting point.

NO.

$$\tilde{v}_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$\tilde{v}_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\tilde{v}_3 = \frac{v_3}{\|v_3\|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\Rightarrow \{\tilde{v}_1, \tilde{v}_2, \tilde{v}_3\}$ is an orthonormal set.

Theorem: Let $\{\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n\}$ be an orthogonal set of nonzero vectors.

1. $\{a_1\mathbf{v}_1, a_2\mathbf{v}_2, ..., a_n\mathbf{v}_n\}$ is also orthogonal for any $a_i \neq 0$ in $\mathbb{R}$.
2. $\left\{\frac{1}{||\mathbf{v}_1||}\mathbf{v}_1, \frac{1}{||\mathbf{v}_2||}\mathbf{v}_2, \ldots, \frac{1}{||\mathbf{v}_n||}\mathbf{v}_n\right\}$ is an orthonormal set.

Theorem: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space, then S is linearly independent.
Proof:

Let

$$a_1\vec{v}_1 + \cdots + a_n\vec{v}_n = \vec{0}$$

$$\Rightarrow \langle a_1\vec{v}_1 + \cdots + a_n\vec{v}_n, \vec{v}_1 \rangle = 0$$

$$\Rightarrow a_1\langle \vec{v}_1, \vec{v}_1 \rangle = 0 \Rightarrow a_1 = 0.$$

Similarly, we also have $a_i = 0$ for $\forall\ 1 \leq i \leq n$.

So $\{v_1, \cdots, v_n\}$ is linearly indep.

Remark: Is the converse of this theorem true?

NO.

Theorem: (Pythagoras' Theorem) If $\{\mathbf{f}_1, \mathbf{f}_2, ..., \mathbf{f}_n\}$ is an orthogonal set of vectors, then

$$||\mathbf{f}_1 + \mathbf{f}_2 + \cdots + \mathbf{f}_n||^2 = ||\mathbf{f}_1||^2 + ||\mathbf{f}_2||^2 + \cdots + ||\mathbf{f}_n||^2$$

Theorem: Let $S = \{\mathbf{f}_1, \ldots, \mathbf{f}_n\}$ be an orthogonal basis for an inner product space V and let $\mathbf{v}$ be any vector in V. Then

$$\mathbf{v} = c_1 f_1 + c_2 f_2 + \cdots + c_n f_n$$

where

$$c_i = \frac{\langle v, f_i \rangle}{\langle f_i, f_i \rangle}$$

Example: Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthonormal basis for $\mathbb{R}^3$ (using the dot product) where

$$\mathbf{u}_1 = \begin{bmatrix} \frac{2}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \\ -\frac{2}{3} \end{bmatrix}, \text{ and } \mathbf{u}_3 = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}.$$

Write the vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ as a linear combination of the vectors in S.

$$v = \langle v, u_1 \rangle u_1 + \langle v, u_2 \rangle u_2 + \langle v, u_3 \rangle u_3$$

$$= u_1 + 7u_3.$$

The Gram–Schmidt Orthogonalization Algorithm

Theorem: Let V be an inner product space and $W \neq \{\emptyset\}$ be an m-dimensional subspace of V. Then there exists an orthonormal basis $T = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ for W.

1. Start with a basis $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ for W.
2. Set $\mathbf{v}_1 = \mathbf{u}_1$.
3. Set $\mathbf{v}_2 = \mathbf{u}_2 - \dfrac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1$. The vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ will be orthogonal**. *Optional:* Multiply $\mathbf{v}_2$ by a scalar to clear fractions.
4. Set $\mathbf{v}_3 = \mathbf{u}_3 - \dfrac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \dfrac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2$. *Optional:* Multiply $\mathbf{v}_3$ by a scalar to clear fractions.
5. Continue on in this way, so
$$\mathbf{v}_k = \mathbf{u}_k - \sum_{i=1}^{k-1} \frac{\langle \mathbf{u}_k, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2}\mathbf{v}_i.$$
Optional: Multiply each $\mathbf{v}_i$ by a scalar as you find it to clear fractions.
6. The resulting set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ is an orthogonal basis for W.
7. Now let $\mathbf{w}_i = \dfrac{1}{||\mathbf{v}_i||}\mathbf{v}_i$ for $i = 1, 2, \ldots, m$. The set $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ is an orthonormal basis for W.

**Why are $\mathbf{v}_1$ and $\mathbf{v}_2 = \mathbf{u}_2 - \dfrac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1$ orthogonal?

Take $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle =$ $\left\langle v_1 , u_2 - \dfrac{\langle u_2, v_1\rangle}{\|v_1\|^2} \cdot v_1 \right\rangle$

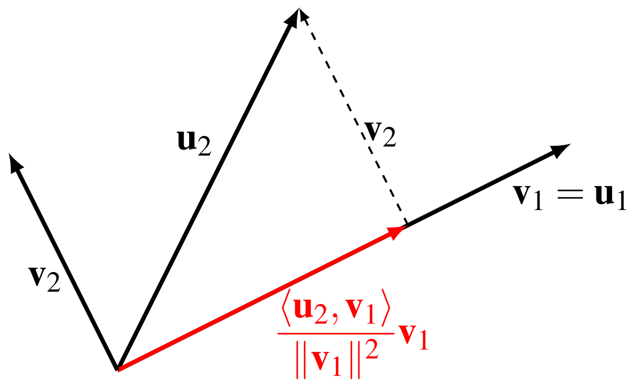
$= \langle v_1 , u_2 \rangle - \langle u_2 , v_1 \rangle = 0 .$

Okay, but seriously, what are we doing?

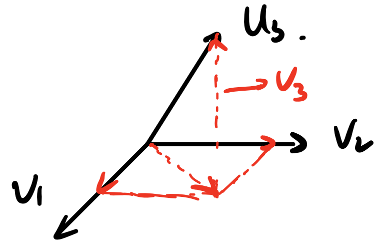
If we think of $\langle \mathbf{u}, \mathbf{v} \rangle$ as dot product, then

$$\frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 = \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 = \text{proj}_{\mathbf{v}_1}\mathbf{u_2}$$

So $\mathbf{v}_2 = \mathbf{u}_2 - \text{proj}_{\mathbf{v}_1}\mathbf{u_2}$ creates a vector orthogonal to $\mathbf{v}_1 = \mathbf{u}_1$.



What does the next step $\mathbf{v}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2$ mean geometrically?



Example: Let W be the subspace of $\mathbb{R}^4$ with the standard inner product whose basis is $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, where

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \text{ and } \mathbf{u}_3 = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Transform S into an orthonormal basis $T = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ for W (on the next page).

Space for example:

$$v_1 = u_1$$

$$v_2 = u_2 - \frac{u_2 \cdot v_1}{\|v_1\|^2} \cdot v_1$$

$$= u_2 - \frac{-1}{3} v_1 = u_2 + \frac{1}{3} v_1 = \begin{bmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ 1 \end{bmatrix}$$

$$v_3 = u_3 - \frac{u_3 \cdot v_2}{\|v_2\|^2} \cdot v_2 - \frac{u_3 \cdot v_1}{\|v_1\|^2} \cdot v_1$$

$$= u_3 - \frac{\frac{4}{3}}{\frac{15}{9}} v_2 - \frac{-2}{3} \cdot v_1$$

$$= u_3 - \frac{4}{5} v_2 + \frac{2}{3} v_1$$

$$= \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} - \frac{1}{15} \begin{bmatrix} -8 \\ 4 \\ 4 \\ 12 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} \\ \frac{2}{5} \\ -\frac{3}{5} \\ \frac{1}{5} \end{bmatrix}$$

$$\Rightarrow w_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad w_2 = \frac{1}{\sqrt{15}} \begin{bmatrix} -2 \\ 1 \\ 1 \\ 3 \end{bmatrix}, \quad w_3 = \frac{1}{\sqrt{15}} \begin{bmatrix} 1 \\ 2 \\ -3 \\ 1 \end{bmatrix}$$

Remark: The Gram–Schmidt orthogonalization is possible in all inner product spaces, not just $\mathbb{R}^n$ and its subspaces.